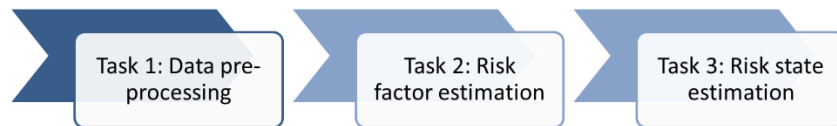


Three tasks are planned for this Final Project Assignment, as shown below. Task 2 and 3 can be done in parallel.



Task 1

Perform data pre-processing activities. This may include data preparation, data cleaning, normalization, standardization, and feature analysis.

Task 2

Develop a model that allows you to estimate the risk factor (R) for new unseen operational scenarios. Provide arguments for the choice and design of your final model(s). Provide arguments for your final model(s) assessment and validation procedure.

Task 3

Develop a model that allows you to estimate the risk state (class) for new unseen operational scenarios. Provide arguments for the choice and design of your final model(s). Provide arguments for your final model(s) assessment and validation procedure.

Other instructions

- You will work in pairs.
- Decisions need to be made together, but Tasks can be done individually.
- We recommend splitting the tasks. Any member must be capable of arguing any decision made.
- At least one of the models must be a *deep neural network* (Lecture 5).
- One report per pair. The report must follow the proposed structure with a maximum number of pages of 10.
- Deadline: Week 8.

Deliverables

1. Final Project Report (see instructions below)
2. Project Assignment Python code

Report Structure

- Members, emails, student numbers.
- Summary (less than 200 words)
- Detailed ML pipeline (include workflow figure).
- Task 1: selected options, argumentation for the selection, model(s) developed, results, validation, comparisons.
- Task 2: argumentations for the model(s) developed, validations, results, comparisons.
- Task 3: argumentations for the model(s) developed, validations, results, comparisons.
- Conclusions (less than 200 words)

Assessment Criteria

You will be evaluated based on a predefined rubric. Check the course Brightspace page to get access to the rubric.

The Project Final Report can be considered *inadmissible*, which will render a FAIL grade for the group, if

- English is not understandable (e.g., full of typos).
- Deep neural networks were not used (as one of the tested models).
- Figures are not legible.
- The report does not follow the proposed structure.

If the report is considered *admissible*:

- English will *not* render extra points.
- Quality of the Python code will *not* render extra points.

Submission Instructions

Please submit your Final Project Report in a PDF format and your Python code in Brightspace before the deadline.